

BATAM HANG NADIM, IndonesiaWMO: **960870**Lat: **1.1192N**Lon: **104.1136E**Elev: **79**StdP: **14.65**Time Zone: **7.00 (E07)**Period: **02-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	73.9	74.8	68.4	104.6	82.0	69.6	109.1	82.1	23.5	85.5	22.4	85.4	3.9	250	0.364

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	10.4	90.9	78.4	89.9	78.6	89.4	78.6	80.9	86.7	80.4	86.2	80.0	85.7	9.7	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.2	152.0	83.2	79.0	151.0	83.0	78.7	149.2	82.8	44.6	87.4	44.1	86.8	43.7	86.2	85.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.0	18.0	15.6	70.8	93.8	3.5	1.9	68.3	95.2	66.3	96.3	64.4	97.4	61.8	98.8		
(5)			69.3	82.4	2.9	1.2	67.1	83.2	65.4	83.9	63.8	84.5	61.6	85.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	82.2	81.1	81.7	82.5	83.1	83.0	82.9	82.2	82.3	82.1	82.3	81.6	81.2
	DBStd	1.97	2.15	1.74	1.98	1.76	1.68	1.96	1.98	2.00	1.83	1.66	1.58	1.96
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11741	965	888	1009	992	1024	986	1000	1000	962	1000	947	968
	CDD65	6266	500	468	544	542	559	536	535	535	512	535	497	503
	CDH74	65627	4799	4777	5928	5993	6185	5882	5702	5690	5323	5538	4940	4870
	CDH80	20338	1203	1315	1952	2098	2122	2016	1835	1798	1683	1740	1358	1218
Wind	WSAvg	5.8	8.5	9.4	7.3	4.8	4.3	4.8	5.8	6.0	5.2	3.9	3.9	5.9
Precipitation	PrecAvg	95.2	9.7	4.4	7.2	7.5	7.3	6.3	6.8	7.2	6.7	8.5	11.1	12.8
	PrecMax	120.8	25.5	12.9	16.6	12.4	13.6	10.4	10.0	15.8	10.7	14.9	18.2	36.2
	PrecMin	56.5	0.2	0.6	1.5	3.6	2.2	3.4	2.4	3.8	1.9	2.4	5.2	4.8
	PrecStd	17.1	8.1	3.5	4.1	2.5	2.6	2.4	2.0	2.7	2.0	3.7	3.7	7.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	89.2	89.1	90.5	91.7	91.7	91.5	90.0	90.0	90.2	91.3	89.9	89.5
		MCWB	78.0	77.6	78.1	78.8	79.1	77.9	78.7	78.1	77.9	76.9	78.2	77.8
	2%	DB	87.7	87.9	89.2	90.1	90.1	90.0	89.1	89.0	89.2	89.7	88.5	87.8
		MCWB	77.9	77.5	78.0	79.0	79.3	79.0	78.6	78.2	78.0	77.7	78.2	77.8
	5%	DB	86.4	87.0	88.1	89.2	89.3	89.1	88.1	88.0	88.2	88.4	87.5	86.5
		MCWB	77.4	77.2	78.0	79.0	79.4	79.2	78.6	78.2	78.1	78.1	78.2	77.8
	10%	DB	85.6	86.0	87.2	87.9	88.0	87.9	87.2	87.1	87.4	87.3	86.2	85.5
		MCWB	77.0	76.9	77.8	78.8	79.3	79.1	78.5	78.1	78.1	78.1	78.2	77.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	79.7	79.3	80.1	81.0	81.7	81.1	80.9	80.1	80.2	80.1	80.1	79.9
		MCDB	85.8	85.7	86.8	87.3	87.4	87.3	86.5	85.9	86.2	85.6	85.4	85.3
	2%	WB	79.0	78.8	79.2	80.4	80.9	80.7	80.0	79.7	79.6	79.6	79.4	79.1
		MCDB	84.6	85.1	85.7	86.6	86.5	86.7	85.8	85.4	85.5	85.1	84.6	84.4
	5%	WB	78.4	78.2	78.9	79.9	80.3	80.1	79.6	79.1	79.1	79.1	79.0	78.7
		MCDB	83.7	84.2	85.3	85.8	85.7	85.8	85.1	84.7	84.9	84.6	84.2	83.8
	10%	WB	77.9	77.7	78.4	79.5	79.9	79.7	79.1	78.8	78.7	78.7	78.6	78.2
		MCDB	83.2	83.5	84.5	85.2	85.0	85.2	84.5	84.3	84.4	84.1	83.6	83.0
Mean Daily Temperature Range	5% DB	MDBR	8.4	8.9	9.5	10.5	10.4	10.2	9.7	9.7	10.5	10.6	9.8	8.6
		MCDBR	9.7	9.7	10.3	11.7	11.6	11.4	10.7	10.5	11.7	12.1	11.6	10.1
		MCWBR	3.0	2.8	2.9	3.6	3.7	3.6	3.4	3.4	3.8	3.8	3.5	3.2
	5% WB	MCDBR	9.1	9.1	9.9	10.9	10.7	10.6	10.0	9.9	10.9	10.9	10.2	9.3
		MCWBR	3.2	3.1	3.3	3.9	4.0	3.9	3.8	3.8	4.1	4.0	3.8	3.5
Clear-Sky Solar Irradiance	taub	0.470	0.481	0.486	0.475	0.463	0.473	0.467	0.510	0.543	0.571	0.480	0.470	
	taud	2.308	2.260	2.253	2.284	2.331	2.304	2.327	2.181	2.082	1.985	2.273	2.316	
	Ebn at Noon	273	272	269	266	262	255	259	253	250	245	268	271	
	Edh at Noon	43	46	46	43	40	40	40	48	54	59	44	42	
All-Sky Solar Radiation	RadAvg	1452	1688	1676	1613	1489	1407	1403	1437	1440	1422	1325	1305	
	RadStd	121	149	142	102	74	101	115	89	93	105	87	102	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (7 neighbors)	+0.67	+0.16	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A